

4 (long) An FET is used in an amplifier circuit as shown in Fig. 4(a). The small signal parameters of the FET are $g_m = 5 \text{ mS}$ and $r_D = 15 \text{ k}\Omega$. The $R_S = 6 \text{ k}\Omega$ and $R_G = 2 \text{ M}\Omega$.

(a) Calculate the gain and the output impedance of the circuit. [12]

(b) Due to electrical interference, noise in the form of a voltage with 200 Hz frequency is induced in the drain of the FET. This noise can be represented by including a small voltage source v_N at the drain, as shown in Fig. 4(b). Draw the small signal equivalent circuit and by setting $v_{in}=0$, derive the expression for the contribution of this noise to the circuit's output voltage. [12]

(c) Determine the maximum amplitude of v_N if the noise contribution to the output must not exceed $30 \text{ }\mu\text{V}$. [6]

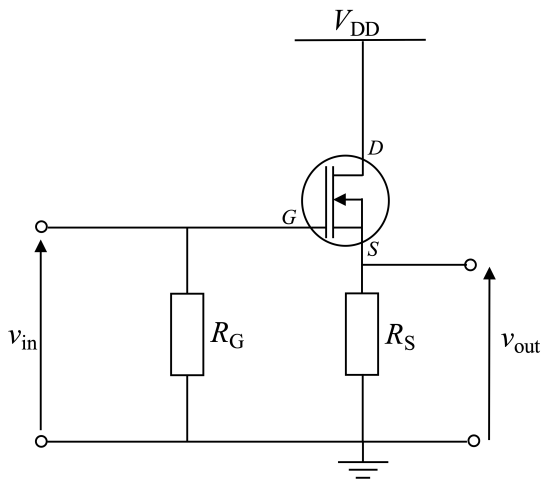


Figure 4 (a)

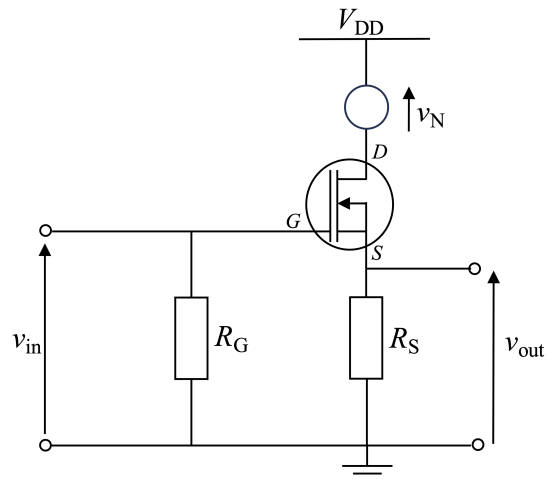


Figure 4 (b)

Fig. 4